z-test (1-sample)

We have to assume normality if n<30.

sigma is known.

alpha: significance level.

1.Hypotesis:

$H_0 : m = m_0$

$H_1 : m \neq m_0$ (2-tailed)

$\quad m < m_0$ (left tailed)

$\quad m > m_0$ (right tailed)

2.Test-statistics:

$$z = \frac{\overline{x} - m_0}{\sigma} \sqrt{n} \sim \mathcal{N}(0,1), \text{if } H_0 \text{ is true.}$$

3.P-value:

The probability of taking a worse test statistic.

$p = 2\phi(-|z|), \quad$ (2-tailed),
$p = \phi(z), \quad$ (left-tailed),
$p = \phi(-z), \quad$ (right-tailed),
where $\phi()$ is the CDF of the standard normal distribution.

4.Decision:

$p > \alpha$: accept H_0

$p \leq \alpha$: reject H_0, accept H_1

Confidence interval:

With confidence level $(1-\alpha) \cdot 100\%$:

$\mathbb{P}(m \in CI_{1-\alpha}) \leq 1 - \alpha$

Example

Create a vector containing the first column of the students' exam grades data.

```
load examgrades
x = grades(:,1);
```

Test the null hypothesis that the data comes from a normal distribution with mean `m = 75` and standard deviation `sigma = 10`.

```
[h,p,ci,zval] = ztest(x,75,10)
```

```
h =
```

```
0
p =
0.9927
ci = 2×1
   73.2191
   76.7975
zval =
0.0091
```

Test the null hypothesis that the data comes from a normal distribution with mean `m = 65` and standard deviation `sigma = 10`, against the alternative hypothesis that the mean is greater than 65.

```
[h,~,~,zval] = ztest(x,65,10,"Tail","right")
```

```
h =
1
zval =
10.9636
```

Plot the standard normal distribution, the returned *z*-statistic, and the critical *z*-value. Calculate the critical *z*-value for the default confidence level of 95% by using `norminv`.

```
k = linspace(-15,15,300);
y = normpdf(k);
zvalpdf = normpdf(zval);
zcrit = norminv(0.95);

plot(k,y);
hold on
scatter(zval,zvalpdf,"filled")
xline(zcrit,"--")
legend(["Standard Normal pdf","z-Statistic", ...
    "Critical Cutoff"])
hold off
```

t-test (1-sample)

sigma is unknown.

Differences from z-test:

2.Test-statistic:

$$t = \frac{\overline{x} - m_0}{s^*}\sqrt{n} \sim t_{n-1}, \text{ if } H_0 \text{ is true.}$$

3.P-value:

Use the CDF of the t_{n-1} distribution.

Example

The daily energy intake in kJ for 11 women are the following:

```
daily_intake = [5260,5470,5640,6180,6390,6515,6805,7515,7515,8230,8770]
```

```
daily_intake = 1×11
        5260        5470        5640        6180        6390        6515 ···
```

```
mean(daily_intake)
```

```
ans = 
6.7536e+03
```

```
std(daily_intake)
```

```
ans = 
1.1421e+03
```

Test the hypotesis that the mean energy intake for women is 7725 kJ.

```
h = ttest(daily_intake,7725)
```

```
h = 
1
```

Make a 99% confidence interval.

```
[~,~,ci] = ttest(daily_intake,7725,"Alpha",0.01)
```

```
ci = 1×2
10^3 ×
    5.6623    7.8450
```

t-test (2-sample)

Shortly after metric units of length were officially introduced in Australia, each of a group of 44 students was asked to guess, to the nearest metre, the width of the lecture hall in which they were sitting. Another group of 69 students in the same room was asked to guess the width in feet, to the nearest foot. The data were collected by Professor T. Lewis and are taken from Hand et al (1994). The main question is whether estimation in feet and in metres gives different results.

Read the 'roomwidth.csv' file into a table.

```
data = readtable('roomwidth.csv');
```

Convert every observations into feet, if 1 meter = 3.28 feet.

```
width = data.width;
unit = categorical(data.unit);
wfeet = width(unit == 'feet');
wmeter = width(unit == 'metres')*3.28;
```

Compare the guesses respect to unit with a boxplot.

```
boxplot(width, unit)
```

Did the students guessed the same way in meters as in feet?

```
[h,p] = ttest2(wfeet, wmeter)
```

```
h =
1
p =
0.0102
```

Exercise.

After the students had made the estimates of the width of the lecture hall the room width was accurately measured and found to be 13.1 metres (43.0 feet). Use this additional information to determine which of the two types of estimates was more precise (ie. which has the higher p-value).

```
%
```

t-test (paired)

```
load examgrades
x = grades(:,1);
y = grades(:,2);
```

Test the null hypothesis that the pairwise difference between data vectors x and y has a mean equal to zero.

```
[h,p] = ttest(x,y)
```

```
h =
0
p =
0.9805
```